

26年嵌赛——英飞凌赛道说明

2026/02

英飞凌大中华区消费、计算与通讯业务事业部



嵌赛英飞凌支持

– 获得技术支持，使用ModusToolBox/Vs code IDE相关问题：

- 英飞凌论坛嵌赛专区：

<https://community.infineon.com/t5/%E5%A4%A7%E5%AD%A6%E7%94%9F%E5%B5%8C%E5%85%A5%E5%BC%8F%E5%A4%A7%E8%B5%9B/bd-p/studentcompetition/page/1>

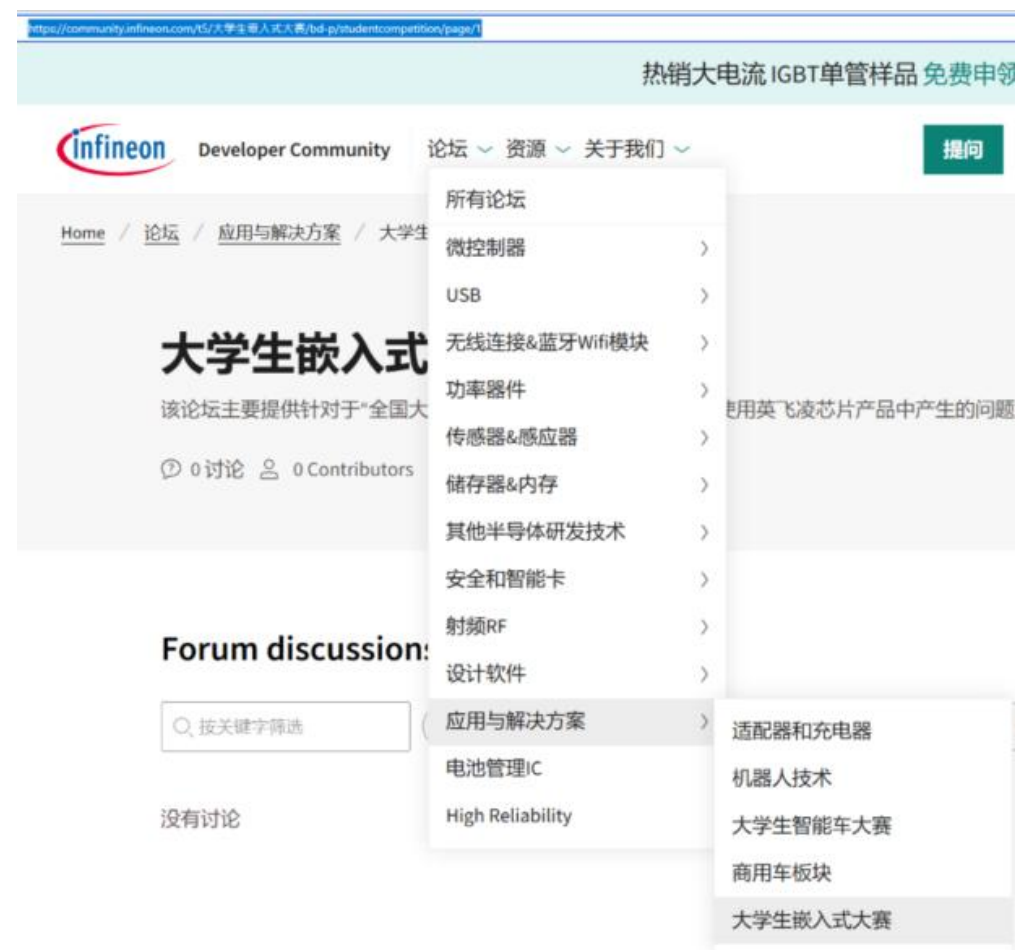
- 英飞凌论坛：<https://community.infineon.com/>

- 21ic：英飞凌MCU论坛：<https://bbs.21ic.com/iclist-59-1.html>

– 使用RT-Thread Studio IDE相关问题：

- RT-Thread Community:

<https://club.rt-thread.org/index.html>



Infineon Company presentation

Infineon is a global leader in power systems and IoT

Global leader

in automotive, power management,
energy efficient technologies and IoT

~57,000

employees¹

Market position

Automotive

#1

TechInsights,
March 2025

Power

#1

Omdia,
October 2025

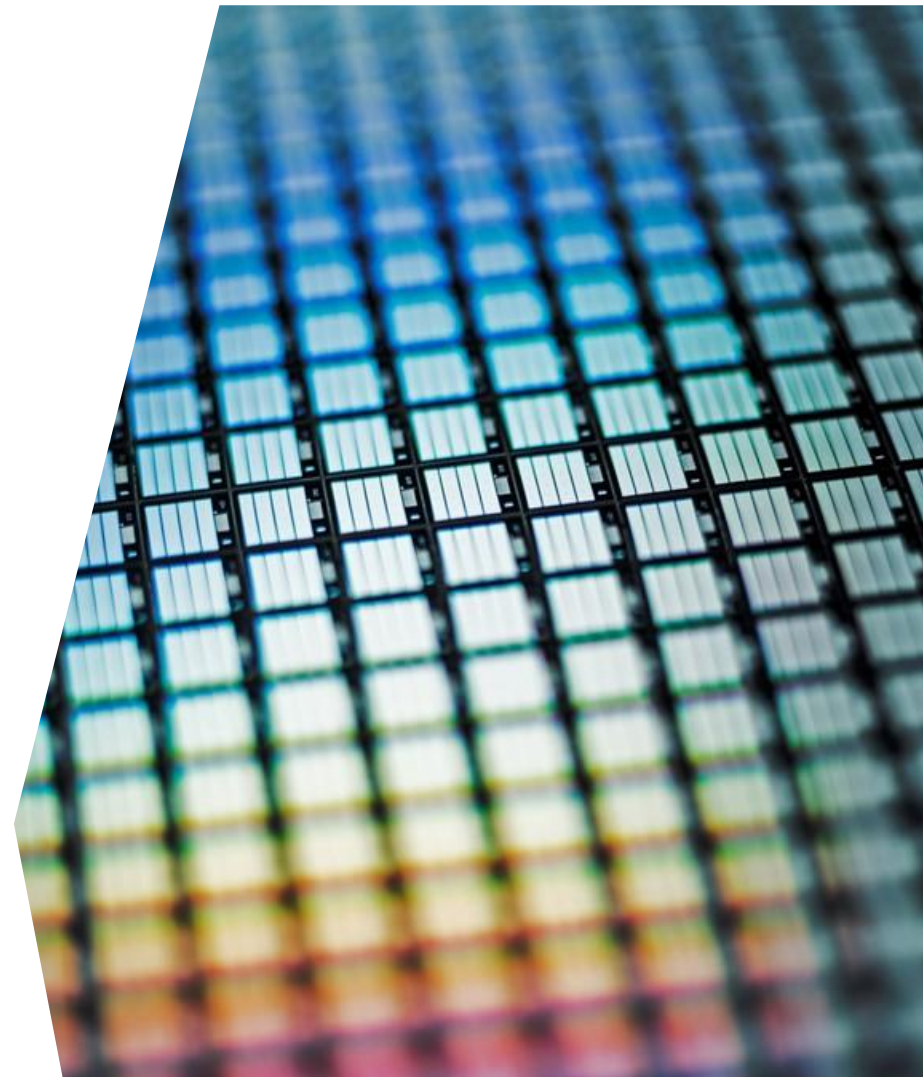
Microcontroller

#1

Omdia,
August 2025



Infineon Official
Wechat



¹ As of 30 September 2025

Infineon at a glance

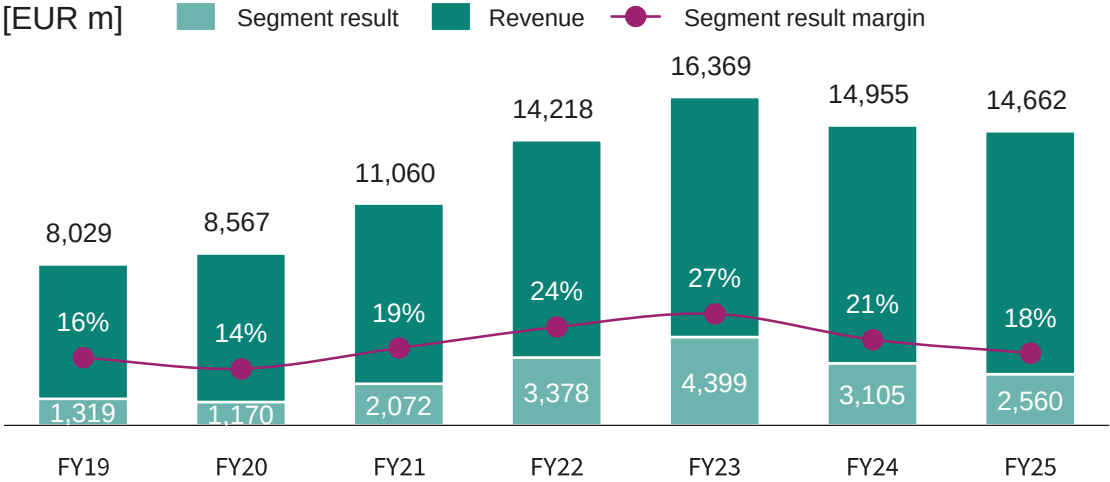
Growth areas

Energy
green and efficient

Mobility
clean and safe

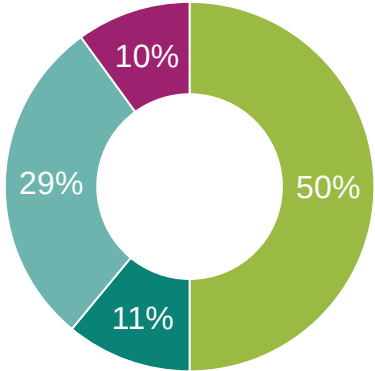
IoT
smart and secure

Financials



FY25 revenue by segment¹

- Automotive (ATV)
- Green Industrial Power (GIP)
- Power & Sensor Systems (PSS)
- Connected Secure Systems (CSS)

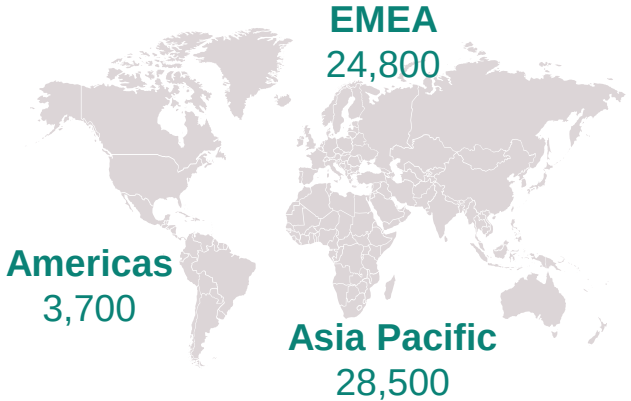


Employees¹

57,000
employees worldwide

75
R&D and

14
manufacturing locations²



For further information: [Infineon Annual Report](#).
¹ 2025 Fiscal year (as of 30 September 2025) | ² As of 30 September 2025

嵌赛用的MCU平台

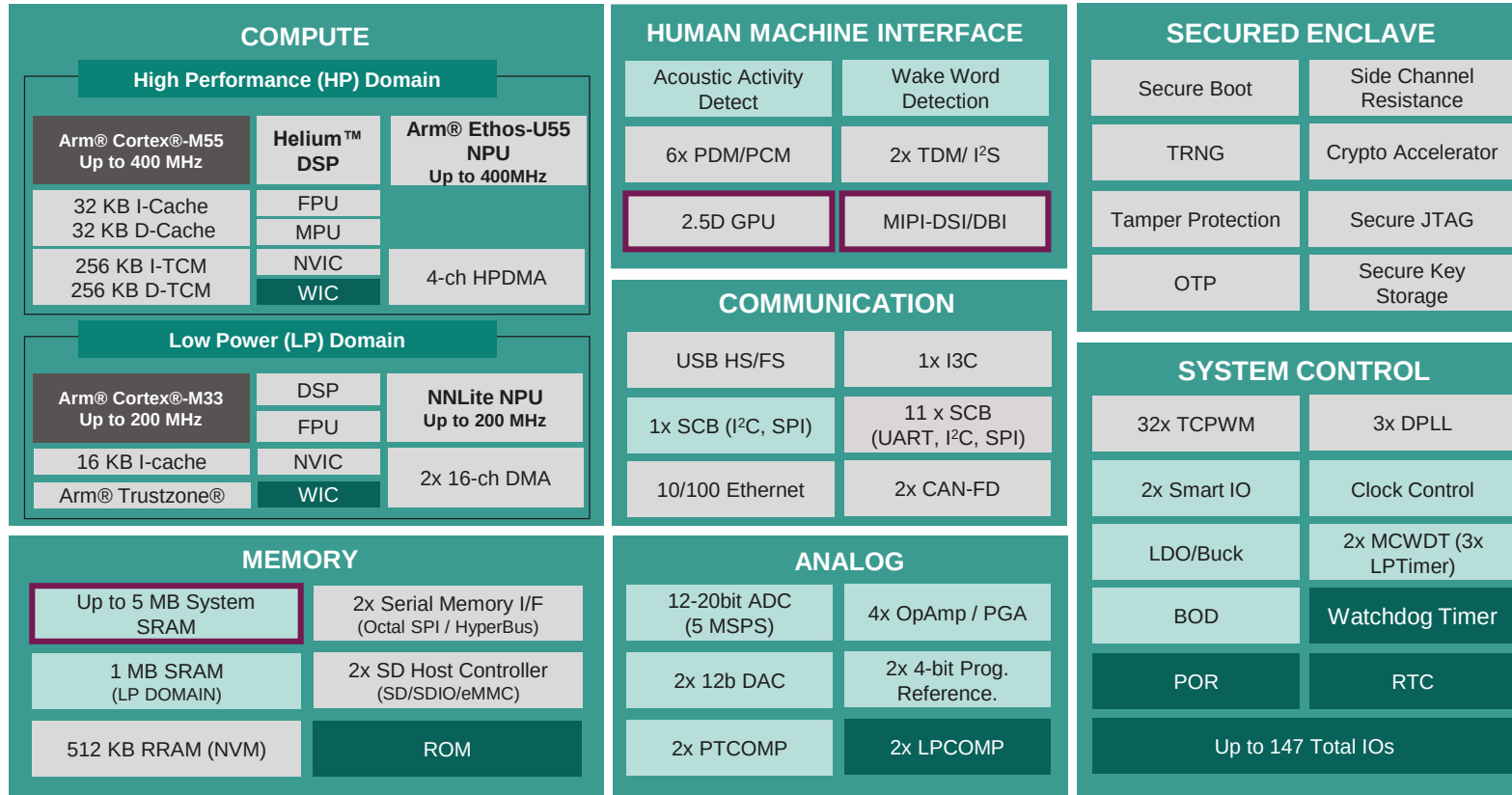
Infinion IoT Compute & Wireless (ICW) Microcontroller Portfolio Overview



	Entry-level MCUs 32-bit Arm® Cortex® M0/M0+ (32–384KB Flash)	Mid-to High-Performance MCUs 32-bit Arm® Cortex® M33/M4/M7/M55/M85 (128KB–4MB Flash)	
Consumer/IoT MCUs	Flexible PSOC™ 4 MCUs with M0/M0+, analog sensor integration, CAPSENSE™ capacitive touch, inductive sensing, wired and wireless connectivity such as USB, CAN, and BLE	Ultra-low power PSOC™ 61 & 62 MCUs with M4/M0+ dual-core and EPC security, ideal for battery powered applications	Next-gen of main-line IoT/Consumer MCUs PSE84: M55+U55+M33+NNLite EPC2/4 Now
Connected MCUs	PSOC™ 4 MCU with AIROC™ Bluetooth LE for IoT applications where ultra low sleep current, CAPSENSE™, analog IOs for sensor data acquisition along with BLE are required	PSOC™ 63 MCU with AIROC™ Bluetooth LE and M4/M0+ dual-core for low-power IoT devices, optimized for AI/ML Edge 2027	AIROC™ and PSOC™ Connect Bluetooth® MCUs AIROC™ Wi-Fi 6/6E+BLE Combo MCU with M33 for ultra-low power with Wi-Fi Range Boost; Matter over Wi-Fi 2027
Industrial MCUs	XMC™ 1000 entry-level MCUs with M0 for industrial applications like power tools, LED lightning, eBike and fan motor control	XMC™ 4000 MCUs with M4F, built-in DSP instruction set, designed for digital power conversion, motor control, sense & control, and IO applications. XMC™ 4300 and 4800 with integrated EtherCAT® XMC™ 5000 with M4F for industrial applications, 5V high pin count and functional safety XMC™ 7000 low-power MCUs with single- or dual-core M7 are built on 40-nm process technology addressing high-end industrial applications	PSOC™ Control C3 MCUs with M33 single- and multi-core, real-time control with high speed control loops, analog and timers. SiC/GaN ready (high switching speeds), up to PSA L3/EPC3 security PSOC™ Control C4 MCUs 2027

Block diagram

System Power Modes: Active/Sleep DeepSleep Hibernate



Product Highlights

- **High performance, real-time compute domain:**
 - Cortex®-M55 w FPU + Helium™ DSP + Ethos-U55 for ML
 - **Up to 5 MB System SRAM**, 256 KB I&D TCMs
- **Low power, real-time compute domain:**
 - Cortex®-M33 and DSP + IFX NNLite for ML
 - 512 KB RRAM, 1 MB SRAM
- **ML:** Advanced ML leveraging Ethos-U55 and NNLite
- **HMI:**
 - Traditional MCU HMI
 - Local voice, cloud voice
 - Vision for friction free interface & safety
 - **Low power Graphics, up to 1024x768, MIPI-DSI/DBI**
- **Peripherals & IO:**
 - USB, 10/100 Ethernet, CAN, SPI, UART, I2C, I3C, I2S
 - Ultra-low-power always-on analog
- **Security:**
 - Secured Enclave @ 25 JIL pts, Infineon Edge Protect Category 4 (EPC4) with PSA certified L3/L4 iSE
 - Edge-Protect Bootloader and TF-M stack available in ModusToolbox™

Status/Availability

- In Development, prelim. DS available
- Samples & EVK available (Edge E84)
- Qualified samples Q2 2025
- SOP: Oct 2025

Packages

WLB154 (5.2x4.3x0.5mm, 0.35mm)	BGA220 (10x10x1.2mm, 0.65mm)	eWLB235 (7x7x0.7mm, 0.4mm)
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Operating Range

-20 to 70°C Ta (Consumer), -40 to 105°C Ta (Industrial)

Applications and Target Markets

- Smart Home, Appliances, Residential AC, Wearables, Industrial HMI, Smart Speakers, etc.

Operating Range

- -20 to 70°C Ta (Consumer)
- -40 to 105°C Ta (Industrial)

PSoC™ 6 MCU – CY8C624ALQI_S2D42 overview

Features

MCU Subsystem

- Dual-core architecture: 150-MHz Arm® Cortex®-M4 and 100-MHz Arm® Cortex®-M0+, DMA controllers
- Ultra-low-power (0.9 V) and low-power (1.1 V) operation mode
- **Up to 2048 KB Flash, 1024 KB SRAM.** Option for QSPI external flash for XIP, data storage

Analog Blocks

- 1x 12-bit 2 MSPS SAR ADC, 16 channel hardware sequencer
- 2x Low power comparators
- CAPSENSE™ capacitive-sensing block, Segment LCD drive capability

Digital Blocks and Communication Interfaces

- **I2S and PDM-PCM converter for interfacing to audio codecs, PDM microphones**
- **2x SD Host Controller/eMMC/SD controllers**
- Quad SPI External Memory Interface with the on-the-fly encryption/decryption
- 24 x 16-bit and 8 x 32-bit timer/counter/pulse-width modulation blocks (TCPWM)
- 12 x serial communication blocks (SCBs) – I2C (12) / SPI (12) / UART (8), deep-sleep SCB – I2C / SPI
- USB 2.0 FS (Host and Device)

Security Features

- Advanced cryptographic coprocessor (Crypto) and True random number generator
- One-time programmable eFUSE for secure key storage
- Secure over-the-air (OTA) firmware update with read-while-write Flash technology for firmware updates

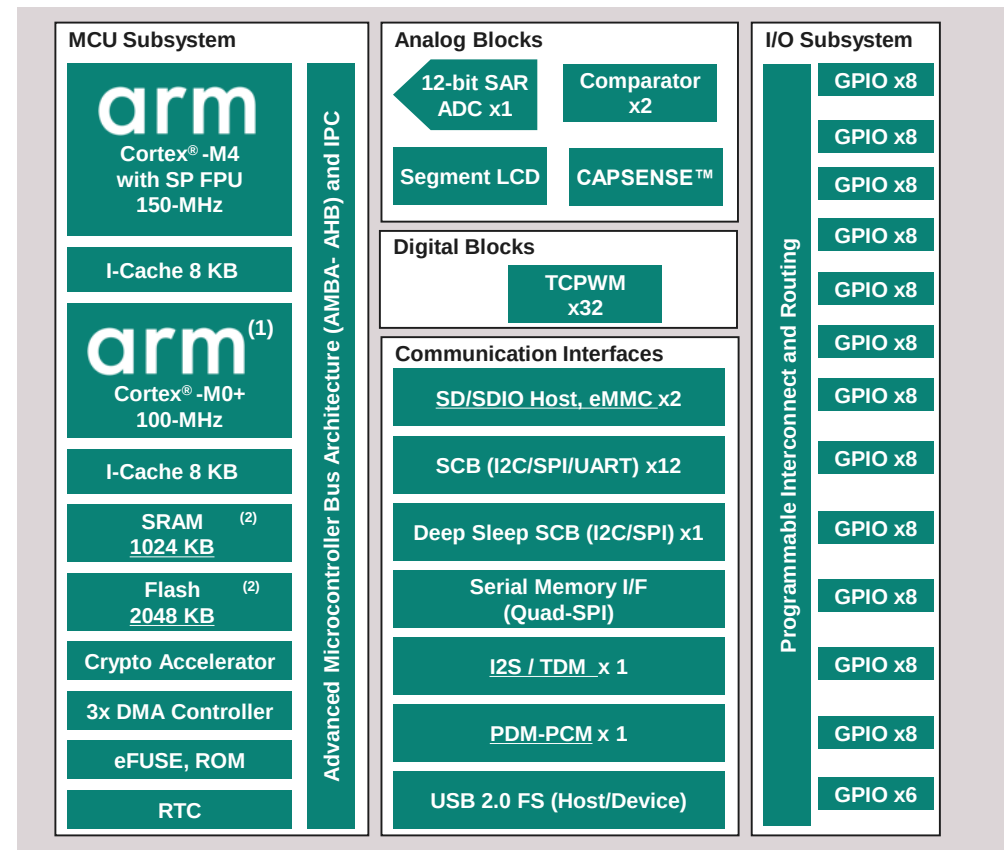
I/O Subsystem: Up to 102 GPIOs

MPN Options

- Packages: 128-TQFP (102 IOs), 124-BGA (100 IOs), 100-WLCSP (82 IOs), 68-QFN (53 IOs)
- Available in the PSoC™ 61 Programmable Line (CY8C61x8, CY8C61xA), PSoC™ 62 Performance Line (CY8C62x8, CY8C62xA), PSoC™ 64 Secured MCU Line (CYB0644A)

Collateral

Documentation: [PSoC™ 6 Documentation Portal](#)



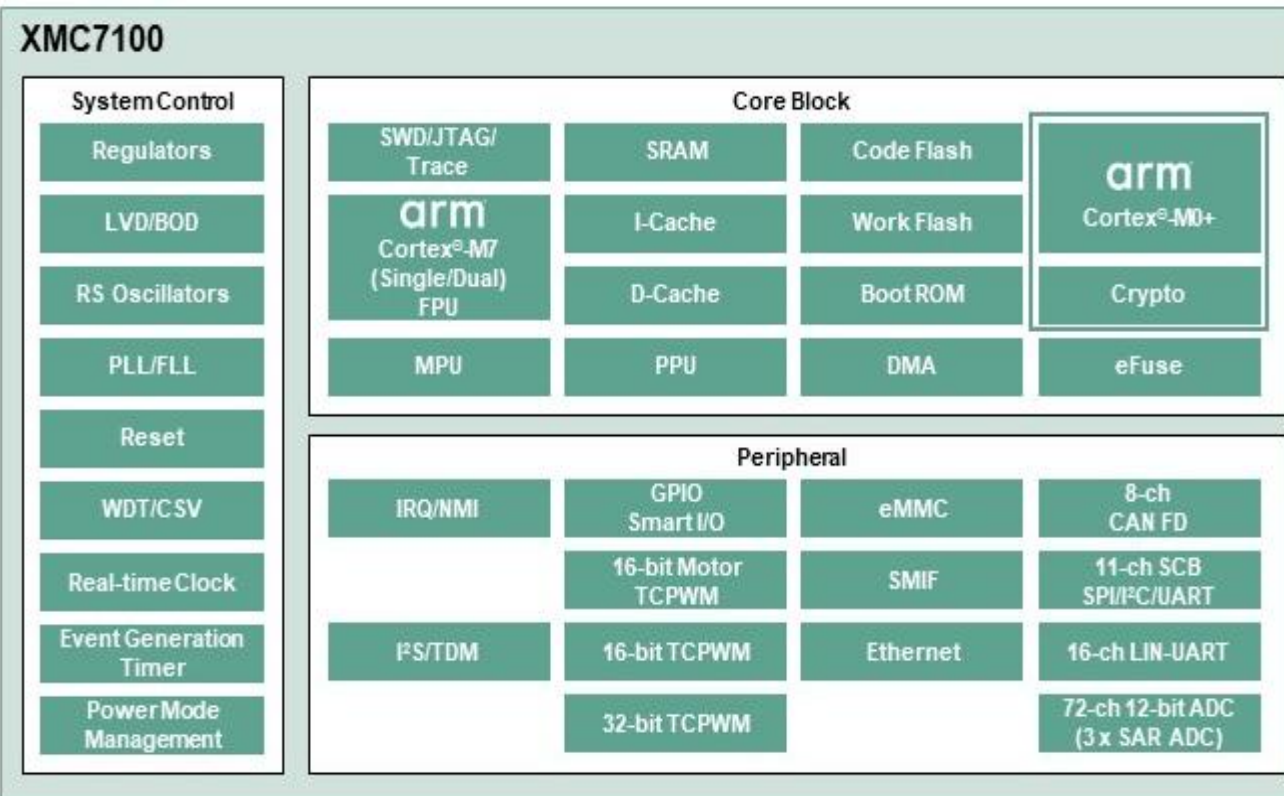
⁽¹⁾ Arm® Cortex®-M0+ CPU not available for user applications in PSoC™ 61 family

⁽²⁾ MPNs also available in the 1024 KB Flash, 512 KB SRAM configuration

Availability

In Production

XMC7100 - XMC7100_F144K2112AA



*Industrial communication protocols to be supported (Q3.24):

- Profinet RT
- EtherNET/IP
- CC-Link
- CANopen

Description

Applications / Target Markets

- › Motor control and drives, inverters/converters (UPS, solar), High-end eBikes, LEVs, eV Chargers
- › As main control unit in demanding, high-end applications

Feature highlights

- › **32-bit MCU Core System**
 - Single 250-MHz Arm® Cortex®-M7 and 100MHz Cortex®-M0+
 - Up to 4-MB Flash, up to 768-kB SRAM, I/D-Cache
- › **2.7-V to 5.5-V** Supply Voltage
- › **Up to 125°C** extended temperature range
- › **Interfaces:**
 - Up to 8-ch CAN FD, up to 11-ch SCB
 - eMMC, SMIF (QSPI/HS-SPI), 1-ch 10/100 Mbit Ethernet*
- › **AD Converter**
 - Up to 72-ch, 12-bit with 3x successive approximation ADC (SAR ADC) units
- › **Timers**
 - Up to 12-ch 16-bit motor control, 63-ch 16-bit timer/counter/pulse-width modulation (TCPWM), and 8-ch 32-bit TCPWM
 - Event Generation Timer



2026英飞凌赛题说明

英飞凌选题与推荐主控说明

选题芯片赛题	PSOC Edge	PSoC 6	XMC 7000	选配的传感器等
嵌入式物联网	✓	✓		CYW Wi-Fi&BT XENSIV sensors
边缘人工智能	✓			CYW Wi-Fi&BT XENSIV sensors
工业应用	✓		✓	CYW Wi-Fi&BT XENSIV sensors
自主命题	✓	✓	✓	CYW Wi-Fi&BT XENSIV sensors

- 英飞凌企业特别奖：两支队伍，价值4000元左右的额外奖品。

嵌入式物联网：包括不限于：IoT，智能穿戴，智能家电，智慧医疗，环境检测...

选题平台跟传感器等

1. 主控平台可以选用PSOC Edge 或者PSoC 6系列：操作系统建议使用RT-Thread，或者FreeRTOS/裸机等均可。
2. 如需无线连接，建议使用AIROC Wireless：CYW Wi-Fi & Bluetooth.
3. 如需传感器，建议在XENSIV sensors中选用，或者其它第三方均可。
4. 对应开发板：PSOC Edge: Edgi Talk; PSoC 62 WITH CAPSNESEN EVALUATION KIT

评定重要指标：

1. 功能复杂度：
 - a. PSOC Edge：MCU功能、NPU功能、DSP功能的结合；嵌入式物联网功能为主，边缘AI运行在NPU为辅助等；外设功能的丰富性，如多传感器：多轴、雷达、显示、无线连接等、环境传感器等；
 - b. PSoC 6: MCU功能的使用：外设功能的丰富性，如多传感器：多轴、雷达、显示、无线连接等、环境传感器等；
2. 作品集成度与美观：作品越接近工程样机越好
3. 功耗/续航能力，实际落地产品的考虑等
4. 作品展示时：丰富性、专业性、实用性。

人工智能：边缘人工智能，云端人工智能与边缘设备结合，应用在IoT，智能穿戴，智能家电，智慧医疗，环境检测等



选题平台跟传感器等

1. 主控平台可以选用PSOC Edge：操作系统建议使用RT-Thread，或者FreeRTOS/裸机等均可。
2. 如需无线连接，建议使用AIROC Wireless：CYW Wi-Fi & Bluetooth.
3. 如需传感器，建议在XENSIV sensors中选用，或者其它第三方均可。
4. 对应开发板：PSOC Edge: Edgi Talk, KIT_PSE84_AI

评定重要指标：

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 - a. PSOC Edge：MCU功能、NPU功能、DSP功能的结合；嵌入式物联网功能为主，边缘AI运行在NPU为辅助等；外设功能的丰富性，如多传感器：多轴、雷达、显示、无线连接等、环境传感器等；
 - b. PSoC 6: MCU功能的使用：外设功能的丰富性，如多传感器：多轴、雷达、显示、无线连接等、环境传感器等；
2. 作品集成度与美观：作品越接近工程样机越好
3. 功耗/续航能力，实际落地产品的考虑等
4. 作品展示时：丰富性、专业性、实用性。

工业应用：如智能小车、飞控、马达控制、数字电源等

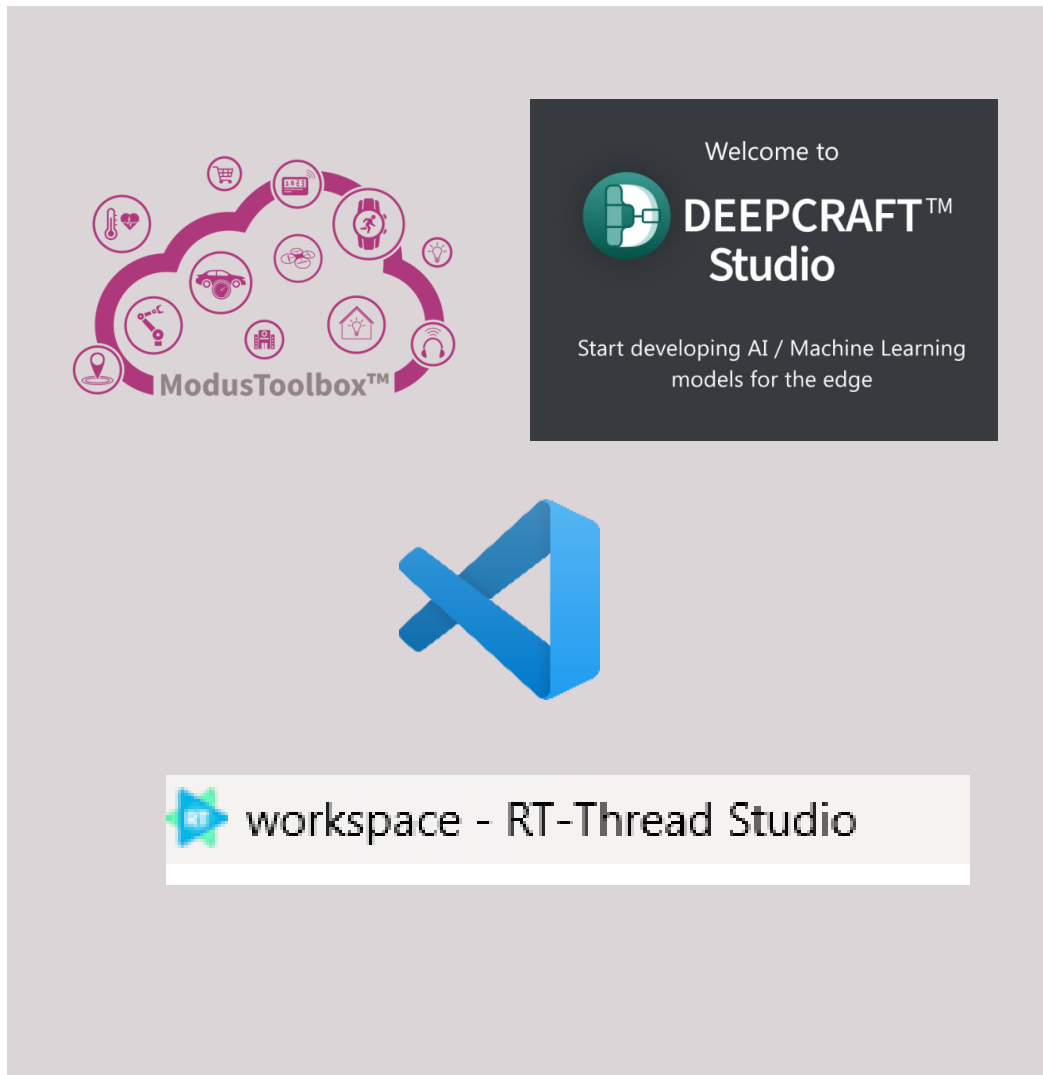
选题平台跟传感器等

1. 主控平台可以选用XMC7100系列，操作系统建议使用RT-Thread，或者FreeRTOS/裸机等均可。
2. 如需无线连接，建议使用AIROC Wireless：CYW Wi-Fi & Bluetooth.
3. 如需传感器，建议在XENSIV sensors中选用，或者其它第三方均可。
4. 对应开发板：XMC7100 kit

评定重要指标：

1. 功能复杂度：
 - a. XMC7100：MCU功能、算法复杂度；外设功能的丰富性，如多传感器：多轴、雷达、显示、无线连接等、环境传感器等；
2. 作品集成度与美观：作品越接近工程样机越好。
3. 应用的性能指标，工业环境干扰下实际落地产品的考虑等。
4. 作品展示时：丰富性、专业性、实用性。

PSoC & XMC IDE



Supported IDEs

- ModusToolBox™: Eclipse IDE, GCC/ARM/LLVM
- DeepCraft Studio: Edge AI data collection/Edge AI model/Label/Model Training/Model convert to c code/sample projects.
- RT-Thread Studio
- Microsoft Visual Studio Code

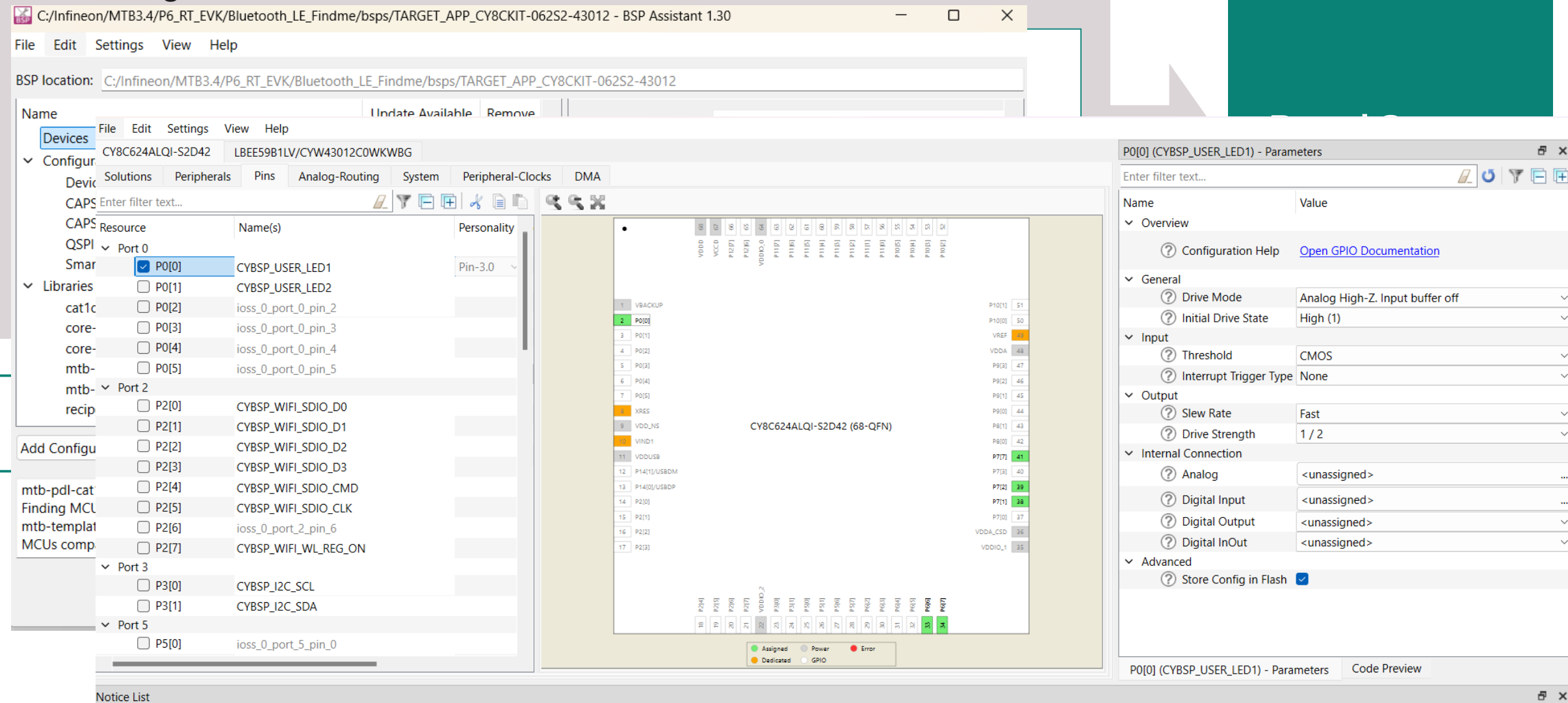
Command-line Interface

- Make based build system with full CLI

Program & Debug kit: PSOC Edge & 6 EVK have the programmer in board; XMC7100 need external J-Link

ModusToolbox™ Software – Tools

BSP Configurators



The screenshot displays the ModusToolbox BSP Assistant 1.30 interface. The main window shows the configuration of the P0[0] pin (CYBSP_USER_LED1) for the CY8C624ALQI-S2D42 device. The left sidebar lists various components and libraries, including the P0[0] pin configuration. The central area shows a pin diagram of the device with P0[0] highlighted. The right sidebar shows the parameters for P0[0] (CYBSP_USER_LED1), including configuration help, general settings, input/output settings, and internal connection settings.

File Edit Settings View Help

BSP location: C:/Infineon/MTB3.4/P6_RT_EVK/Bluetooth_LE_Findme/bsps/TARGET_APP_CY8CKIT-062S2-43012

Devices

File Edit Settings View Help

Update Available Remove

Configurators

Solutions Peripherals Pins Analog-Routing System Peripheral-Clocks DMA

Enter filter text...

Port 0

Resource	Name(s)	Personality
☒	P0[0] CYBSP_USER_LED1	Pin-3.0
☐	P0[1] CYBSP_USER_LED2	
☐	P0[2] ioss_0_port_0_pin_2	
☐	P0[3] ioss_0_port_0_pin_3	
☐	P0[4] ioss_0_port_0_pin_4	
☐	P0[5] ioss_0_port_0_pin_5	

Port 2

☐	P2[0] CYBSP_WIFI_SDIO_D0	
☐	P2[1] CYBSP_WIFI_SDIO_D1	
☐	P2[2] CYBSP_WIFI_SDIO_D2	
☐	P2[3] CYBSP_WIFI_SDIO_D3	
☐	P2[4] CYBSP_WIFI_SDIO_CMD	
☐	P2[5] CYBSP_WIFI_SDIO_CLK	
☐	P2[6] ioss_0_port_2_pin_6	
☐	P2[7] CYBSP_WIFI_WL_REG_ON	

Port 3

☐	P3[0] CYBSP_I2C_SCL	
☐	P3[1] CYBSP_I2C_SDA	

Port 5

☐	P5[0] ioss_0_port_5_pin_0	
--------------------------	---------------------------	--

P0[0] (CYBSP_USER_LED1) - Parameters

Enter filter text...

Overview

Configuration Help [Open GPIO Documentation](#)

General

Drive Mode Analog High-Z. Input buffer off

Initial Drive State High (1)

Input

Threshold CMOS

Interrupt Trigger Type None

Output

Slew Rate Fast

Drive Strength 1 / 2

Internal Connection

Analog <unassigned>

Digital Input <unassigned>

Digital Output <unassigned>

Digital InOut <unassigned>

Advanced

Store Config in Flash ☒

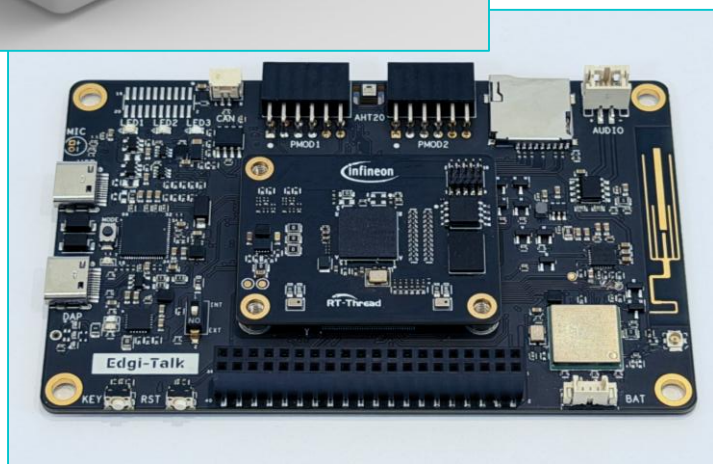
P0[0] (CYBSP_USER_LED1) - Parameters Code Preview

PSOC Edge的开发板 - 26年嵌赛 - 新增全新平台

Edgi Talk EVK: 付押金借用开发板: 赛后归还开发板, 平台退还押金;
或者凭借作品提交记录, 向平台申请退还押金。

链接:

<https://item.taobao.com/item.htm?ft=t&id=1015914924486&skuld=6185934599325>



推荐IDE1: RT-Thread-Studio and Edgi-Talk SDK:

https://github.com/RT-Thread-Studio/sdk-bsp-psoc_e84-edgi-talk

Edgi Talk 资源



推荐IDE2: ModusToolBox -> Vscode + Deepcraft Studio For Edge AI

100+ 例程: Edge AI: Video/Audio/Wi-Fi/BT/Capsense/TFT/ML/Peripherals...

PSoC6的免费开发板

Edgi Talk EVK: 付押金借用开发板: 赛后归还开发板, 平台退还押金;
或者凭借作品提交记录, 向平台申请退还押金。

PSoC 62 WITH CAPSNESEN EVALUATION KIT :

<https://item.taobao.com/item.htm?ft=t&id=894802386964>



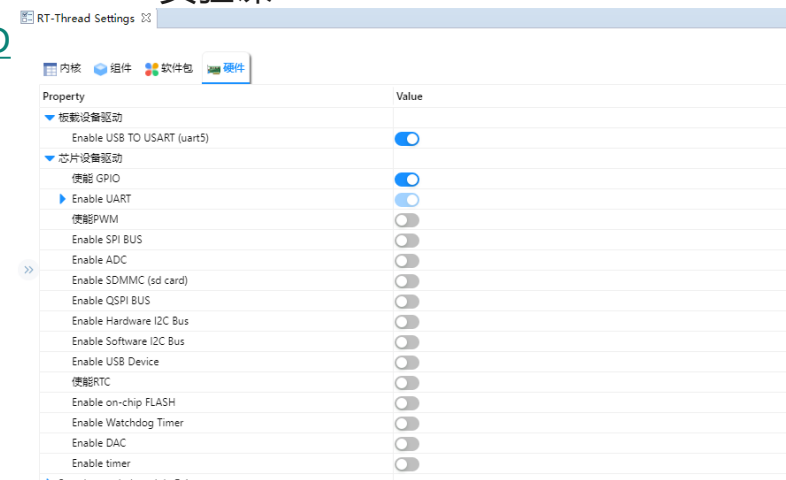
推荐IDE1: RT-Thread-Studio

RT-Thread RTOS + Doc + Git code

<https://github.com/RT-Thread-Studio/sdk-bsp-cy8c624-infineon-evaluationkit>

PSoC 62 EVK with RT-Thread 实验课:

<https://b23.tv/t4id8MD>



推荐IDE2: ModusToolBox

100+ 例程: Wi-Fi/BT/Capsense/TFT/ML/Peripherals...

MTB下修改BSP技术链接:

<https://bbs.21ic.com/icview-3356478-1-1.html>

<https://bbs.21ic.com/icview-3341466-1-1.html>

PSoC62 课程:

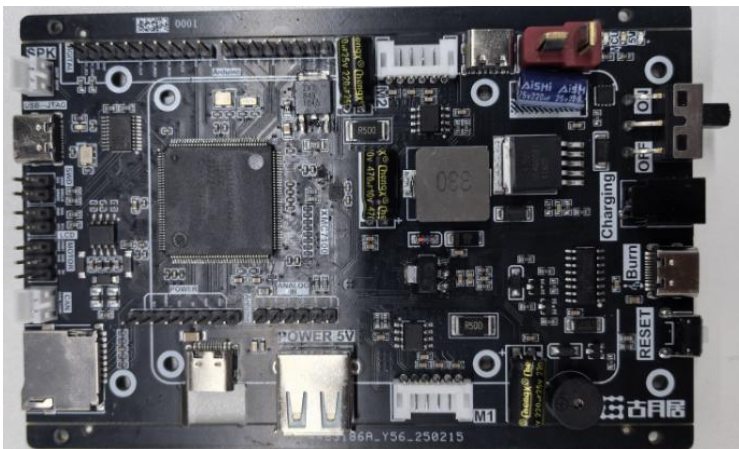
<https://mooc1.chaoxing.com/course-ans/courseportal/259566736.html>

XMC7100的免费开发板

Edgi Talk EVK: 付押金借用开发板: 赛后归还开发板, 平台退还押金; 或者
凭借作品提交记录, 向平台申请退还押金。

XMC7000系列: XMC7100 EVK:

<https://item.taobao.com/item.htm?ft=t&id=896196644440>

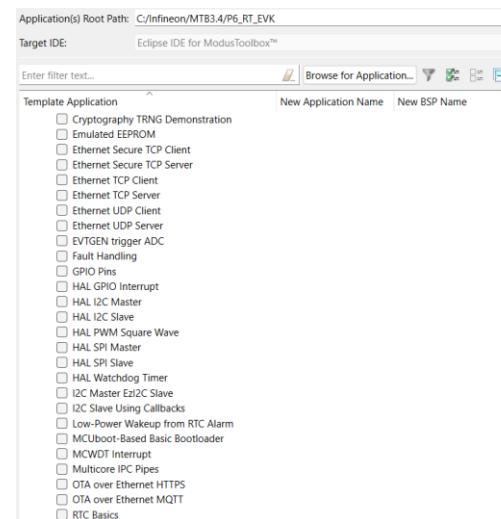


推荐IDE1: ModusToolBox

50+ 例程: CAN/Ethernet/I2C/PWM/ADC/SPI/UART...

技术文档链接:

https://gitee.com/ps-micro/infineon_xmc7100_carrier_board



推荐IDE2: RT-Thread-Studio

<https://github.com/RT-Thread/rt-thread/tree/master/bsp/Infineon/xmc7100d-f144k4160aa>

其它非免费的平台

1. 英飞凌官方天猫旗舰店（如果需要DIY，可自行购买其它配件，但不会进行退费）
https://infineon.world.tmall.com/shop/view_shop.htm?spm=a21n57.shop_search.0.0.7dbc488308636f
2. 如果开发者手头已有基于PSOC Edge/PSoC6/XMC7000主控系列的开发板等，也可以用于参赛并提交作品。
3. 去其它渠道购买到了PSOC Edge/PSoC6/XMC7000芯片，自己DIY板子参赛等都可以。

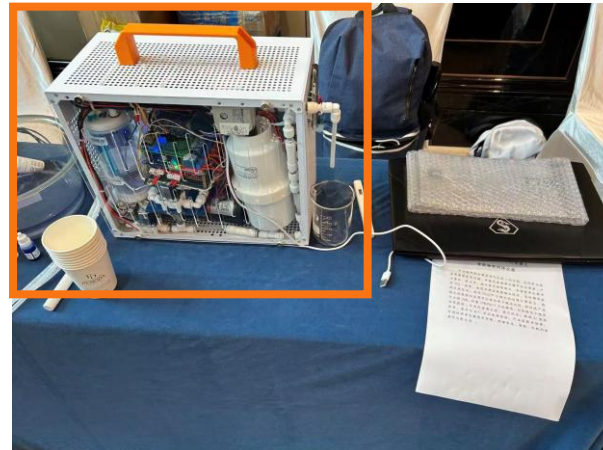
嵌赛英飞凌赛道往年优秀作品

嵌赛英飞凌赛道2025年优秀作品 – 基于PSoC6平台

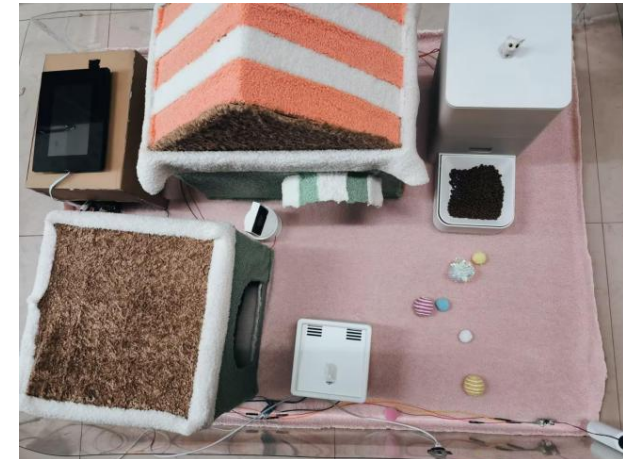
全景智能交互功能小车



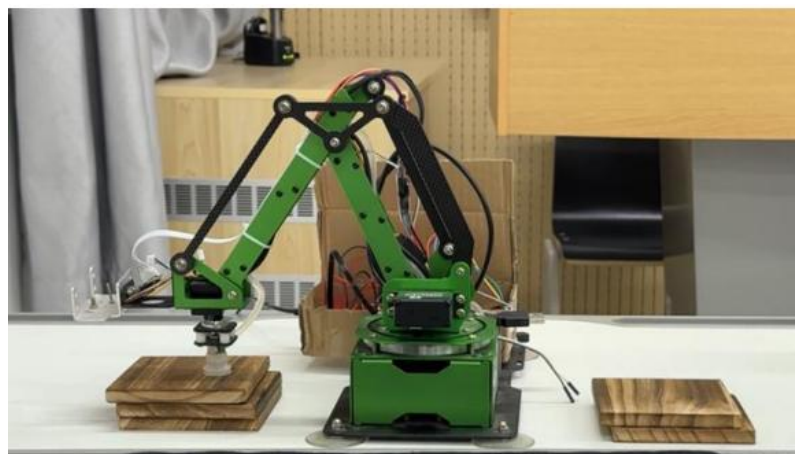
便携式智能物联网净水系统



云猫驿站- 宠物远程监护系统



工业异常检测系统



多模态养老监护系统



嵌赛英飞凌赛道2025年优秀作品 – 基于XMC7000平台

农业巡检机器人



工业异常检测系统



其它

嵌赛英飞凌支持

– 获得技术支持，使用ModusToolBox IDE相关问题：

– 英飞凌论坛嵌赛专区：

<https://community.infineon.com/t5/%E5%A4%A7%E5%AD%A6%E7%94%9F%E5%B5%8C%E5%85%A5%E5%BC%8F%E5%A4%A7%E8%B5%9B/bd-p/studentcompetition/page/1>

– 英飞凌论坛：<https://community.infineon.com/>

– 21ic：英飞凌MCU论坛：<https://bbs.21ic.com/iclist-59-1.html>

– 使用RT-Thread Studio IDE相关问题：

– RT-Thread Community:

<https://club.rt-thread.org/index.html>

